Balanced Orthogonal Subspace Separation Detector for Few-Shot Object Detection in Aerial Imagery

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***Abstract*— Few-shot object detection (FSOD) in remote sensing images (RSIs) aims to achieve object location and classification with only a few training samples. Currently, mainstream transfer-learning methods employ a two-stage approach: pretraining on data-abundant base classes and fine-tuning on few-shot novel classes. However, existing approaches suffer notable degradation in both base and novel classes during fine-tuning, because of gradient conflict and class imbalance. To address this, we construct the balanced orthogonal subspace separation (BOSS) detector, a novel two-stage framework for FSOD. Specifically, to avoid contradictory gradients, BOSS distinctly isolates the training of base and novel classes at both structural and feature levels. For structural separation, a low-rank subspace adapter (LoSA) is introduced to ensure network optimization for novel classes without hampering base classes' pretraining performance, effectively addressing over-fitting in few-shot scenarios. For feature disentanglement, an orthogonal subspace extractor (OSE) is presented, enhancing class separability by learning class-specific, orthogonal basis-spanned subspace. Finally, a balanced classifier (BC) is proposed to equalize the imbalanced loss, with its dual-component design mitigating bias toward predicting background or base classes. Comparative evaluations on diverse remote sensing datasets demonstrate BOSS's superiority, outperforming state-of-the-art methods in mean average precision (mAP). These results underscore BOSS's effectiveness in FSOD, particularly in challenging remote sensing contexts.**

***Index Terms*— Adapter tuning, disentanglement representation, few-shot object detection (FSOD), orthogonal subspace learning, remote sensing images (RSIs).**

I. Introduction

OBJECT detection is a pivotal task in the field of computer vision [1], [2], [3], [4], [5], [6], characterized by its extensive range of practical applications. It plays an instrumental role in enabling intelligent analysis of large-scale remote sensing data. Remote sensing object detection [7], [8], [9], [10], [11], [12], [13], [14] received increasing attention and has evolved alongside the progress in conventional deep learning techniques. Concurrently, the effective training and generalization of these models necessitate abundant datasets and sample annotations. However, in real-world scenarios, certain infrequent categories only contain a limited number of samples. The challenge of adequately acquiring and annotating these ground objects has led to few-shot object detection (FSOD) in aerial imagery as a critical issue.

For natural images, the mainstream FSOD methods include meta-learning [15], [16], [17], [18] and transfer-learning [19], [20], [21]. These methods divide the data into base classes with ample samples and novel classes with few samples, ensuring no overlap exists between them. Meta-learning-based approaches further construct support sets and query sets, where support images from each category are leveraged to identify objects in the query images. Similar to other meta-learning tasks, episodic learning is adopted to facilitate the conversion from base to novel classes. However, such techniques suffer from the drawbacks of intricate data organization, complex pipelines, and low computational efficiency. In contrast, transfer-learning-based research directly implements a two-stage process on the general object detection architecture. This involves thoroughly pretraining on base classes, followed by fine-tuning of selected layers of the model. The primary objective is to preserve the detection capabilities of base classes while effectively transferring their knowledge to novel classes.

In remote sensing scenarios, most recent works employ transfer-learning, capitalizing on the benefits of flexible training, superior performance, and model efficiency. Researchers primarily adapt their methods to accommodate the characteristics of aerial images, such as considering orientations arbitrary and integrating text-modal knowledge. However, transfer-learning comes with an inherent challenge: the optimization gradients conflict between base and novel classes during the fine-tuning stage. Especially concerning various remote sensing objects with high interclass similarity in both object characteristics and scales, the knowledge transfer between different base and novel classes poses a greater difficulty, highlighting the pronounced issue of gradient conflict. The quantitative illustration is provided in Fig. 1. Derived from the gradients of base and novel classes with respect to the network weights, we generate the histogram displaying their gradient direction angle across multiple iterations. Most angle cosine values are notably below 0, indicating obtuse angles

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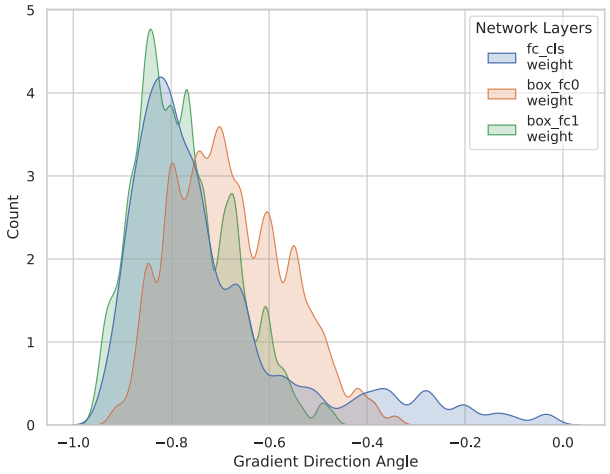


Fig. 1. Conflict gradients in transfer-learning. We divide the loss into base and novel components, calculating the gradient contribution of each to the network on the different layers. The histogram displays the frequency of directional angle cosine between them, where negative values indicate gradients conflict between base and novel classes.

TABLE I
DEGRADATION OF BASE CLASSES

Stage	mAP	Base mAP	Novel mAP
Pretraining	0.549	0.732	-
Fine-tuning	0.507	0.582 (-0.132)	0.282

between the update directions for different losses. This exhibits conflicting effects on network updating, which brings difficulties to parameter optimization. Consequently, for a detector fully pretrained on base classes, maintaining its detection capabilities and generalizing to novel classes depending on few-shot samples are two contradictory tasks. In other words, it is harder for the typical transfer-learning model to extract features from remote sensing objects suitable for both base and novel classes. To achieve effective FSOD, it is imperative to acquire respective representation under separated structures. Even though a few works [22], [23] in natural images attempt to address the base classes degeneration, they have not established a fully isolated framework that considers both features and structures from the perspective of gradient conflicts, thereby impacting the performance of all classes. In fact, this is essential to prevent the network from updating in unfavorable directions. Besides, due to the scarcity of ground truth for novel classes in a few-shot setting, the class imbalance issue arises. Most instances are negative samples labeled as the background. Compared to the base classes with abundant objects, the detection of novel classes in RSIs is inhibited. The results are potentially biased toward predicting background and base classes. Existing methods suffer notable degradation in both base and novel classes during fine-tuning, attributed to the above gradient conflict and class imbalance, as shown in Table I and Fig. 2. After fine-tuning these entangled detectors, it not only greatly reduces the performance in base classes but also influences the learning of novel classes.

Therefore, we propose an innovative detection architecture to avoid gradient conflict and suppress the adverse effects of class imbalance in complex remote sensing contexts, named balanced orthogonal subspace separation (BOSS) detector. It completely isolates the training of base and novel classes at both structural and feature levels, ensuring gradients are unaffected. For structural separation, the low-rank subspace adapter (LoSA) preserves the pretrained model parameters. It modulates the convolutional and linear layers in each component with fewer parameters, alleviating the over-fitting issue in full-parameters fine-tuning under few-shot conditions. For disentangling features in each category, the following orthogonal subspace extractor (OSE) adaptively learns the subspace spanned by orthogonal basis. The disentangled representation based on learned subspace facilitates independence among different remote sensing objects, enhancing separability between base and novel classes. These two modules collectively enable the model to focus on optimizing novel classes during fine-tuning without destroying the performance of base classes. Besides, the balanced classifier (BC) is designed for loss equalization, addressing the class imbalanced sampling. It comprises a base classifier for guided sampling and a novel classifier optimized through equilibrium loss, thereby averting bias toward predicting background or base classes. To depict the performance of our proposed method, we compare it with state-of-the-art FSOD approaches on different remote sensing datasets. All the experimental results suggest that the presented BOSS framework has considerable advantages in all settings, particularly. It maintains detection capabilities in base classes with competitive novel classes performance, significantly improving detection in all categories as depicted in Fig. 2.

Our main contributions are highlighted in the following several aspects.

1) We analyze the inherent issue of gradient conflict and class imbalance in typical transfer-learning methods applied to RSIs. In response, we construct an innovative framework, BOSS, significantly improving the performance of FSOD across both base and novel classes.
2) To prevent gradient conflict, we propose a LoSA and an OSE, distinctly isolating the training of base and novel classes at both structural and feature levels. It enhances the discriminability of novel classes without destroying the performance of base classes.
3) We elaborately design a BC to address the class imbalance. It utilizes pretrained results to sample during fine-tuning and optimizes with equilibrium loss, avoiding bias toward predicting other classes.
4) We compare our BOSS with both meta-learning and transfer-learning approaches, evaluated on diverse remote sensing datasets. The experimental results illustrate BOSS's superiority, outperforming state-of-the-art methods in mean average precision (mAP).

II. RELATED WORK

In this section, we mainly review some research related to object detection and FSOD, including both general and remote sensing fields, along with adapter tuning and orthogonality learning in computer vision tasks.

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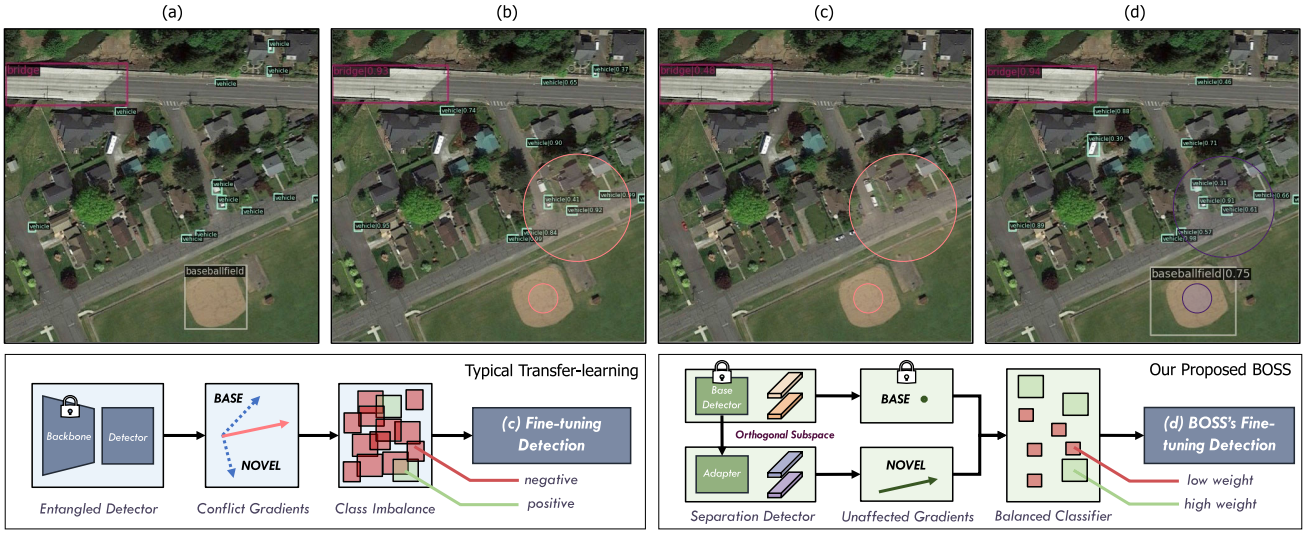


Fig. 2. Illustration of transfer-learning and our BOSS method. For typical transfer-learning, the pretrained model in (b) initially could detect all the base class vehicles, but lost after fine-tuning, and failed to detect the novel class baseball field, indicating poor generalization for novel classes, as highlighted in red in (c). Differently, our proposed BOSS avoids the gradient conflict during the fine-tuning stage by completely isolating the training of base and novel classes at both structural and feature levels. In addition, it suppresses class imbalance resulting from limited annotations for novel classes. Therefore, BOSS prevents base class degradation and focuses on optimizing the knowledge learning of novel classes, as shown in (d). (a) Ground truth. (b) Pretraining. (c) Transfer-learning's fine-tuning. (d) BOSS's fine-tuning.

A. Object Detection

Generic object detection aims to both locate and classify objects, with mainstream research falling into three primary categories: dense prediction detectors, sparse prediction detectors, and two-stage detectors. Dense detectors, exemplified by RetinaNet [2] and FCOS [4], predefine numerous object candidates distributed across the feature map grid using reference points or anchor boxes. These candidates are utilized to predict box elements, including coordinates, width, height, and category. Sparse detectors, represented by works like sparse R-CNN [24], DETR [5], and its variants [25], [26], [27], [28], leverage learnable object queries for set predictions. Two-stage detectors, including the faster R-CNN series [1], [29], [30], [31], incorporate a region proposal network (RPN) to generate region proposals and conduct classification and regression.

Considering the challenges posed by large-scale variations and arbitrary rotations in RSIs, various methods [7], [8], [9], [10], [11], [12], [13], [14] are proposed to adapt the three aforementioned architectures to remote sensing object detection. For instance, oriented RepPoints [32] utilizes point sets based on the dense detector RepPoints [3] to learn object structural representation, improving orientation prediction under dense arrangements. ARS-DETR [33] introduces a transformer-based aspect ratio-sensitive detector, achieving a high-precision measurement.

Although these approaches demonstrate superiority, they depend on large amounts of annotated training samples. Direct application to few-shot settings leads to a significant performance drop. Especially in high-resolution RSIs, complete annotation of objects is highly time-consuming. Therefore, this article focuses on FSOD research, which exhibits stronger applicability to RSIs.

B. Few-Shot Object Detection

Existing FSOD works primarily includes two paradigms: meta-learning [15], [16], [17], [18] and transfer-learning [19], [20], [21]. Meta-learning, represented by meta R-CNN [34], establishes multiple auxiliary tasks to extract knowledge that can be generalized to various detection tasks. This approach facilitates learning to detect objects of each category in query images based on few-shot support images. Nevertheless, the practicality of meta-learning is often constrained by its high computational cost, low efficiency, and complex data structuring. Conversely, transfer-learning methods, such as LSTD [35], TFA [21], and FSCE [20], do not require additional branches for integrating support image information. This leads to more efficient training and inference processes. Initially pretrained on abundant base datasets, these methods acquire adequate base knowledge which is subsequently fine-tuned to adapt to novel classes. It should be highlighted that while some semi-supervised methods may be considered to solve the FSOD problem, they differ significantly. In fact, applying semi-supervised learning to FSOD follows a priori assumption that there are sufficient samples but lacking labels [36]. This constrains the scalability of FSOD for practical applications, where training samples may be scarce. In addition, semi-supervised learning such as consistency regularization, pseudo-labeling [37], generative models, and graph-based models [38] focus on utilizing the relationship between labeled and unlabeled samples within the same class, requiring a small amount of labeled data and a large amount of unlabeled data. This leaves a gap with the FSOD settings, which involves organizing data into base and novel classes, training both with annotations.

In the context of remote sensing, previous FSOD works based on meta-learning [39], [40], [41], [42], [43] incorporate the unique characteristics of RSIs. For instance, FSODM [39] leverages multiscale features of remote sensing objects to devise meta-feature extraction modules. MM-RCNN [41] employs a memory module to store and aggregate category-specific knowledge accrued during training. Recently, some advanced transfer-learning based methods on RSIs [44], [45],

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[46], [47], [48], [49] are proposed. PAMS-Det [44] embeds an involution operator into the backbone, enhancing the classification ability of the features extractor [44]. CIR-FSD [45] utilizes more discriminant context information. Especially, TEMO [48] integrates the latest multimodal model CLIP [50] with transfer-learning based on manually annotated text-modal knowledge to achieve impressive performance through richer priors.

Despite these advances, a critical issue of gradient conflict remains in transfer-learning approaches. A few works in natural images focus on the base classes degeneration caused by this issue, such as retentive RCNN [22] adopting consistency loss to constrain base predictions across different stages, and DiGeo [23] with resampling and maximizing feature distance. However, they lack a direct pursuit of complete separation from the perspective of gradient conflicts, leading to entanglement in feature and structure, respectively. In fact, existing algorithms still need to handle contradictory tasks simultaneously and guarantee optimization. This damages the performance of all categories, especially for RSIs, suffering from the high interclass similarity among diverse objects in aerial images. Furthermore, the numerous candidates labeled as background negative samples in RSIs cause class imbalance. To address these challenges, our BOSS establishes a completely separated detector at both structural and feature levels, incorporating a BC. It avoids gradient conflict and suppresses the adverse effects of class imbalance, demonstrating outstanding transfer capabilities to novel classes without influencing base classes.

C. Adapter Tuning

Adapter tuning [51], [52], [53], [54] is a technique for fine-tuning specific tasks on pretrained language models. It incorporates adapter layers into the model, which are exclusively trained during fine-tuning without altering the parameters of the overall model. This strategy facilitates efficient fine-tuning for particular tasks while preserving the general knowledge captured by large pretrained models. Adapter tuning is subsequently extended to vision foundation models, for fine-tuning vision tasks such as classification [55] and dense prediction [56]. Given that added network layers result in an increase in inference overhead, low-rank adaptation methods [57], [58] are proposed to address this issue. These methods aim to fine-tune the model by introducing a low-rank matrix with fewer parameters into the weight matrix of the model, typically represented as the product of two smaller matrices. This low-rank matrix serves as a modification for the original weight matrix, allowing adjustments during actual computations.

Our method introduces adapter tuning combined with low-rank decomposition into FSOD for the first time. It preserves the pretrained model parameters to extract knowledge from base classes and optimizes the low-rank matrix to learn novel classes, successfully achieving structural separation. In addition, the lightweight characteristics make it well-suited for FSOD, effectively alleviating the over-fitting problem in full-parameters fine-tuning.

D. Orthogonality Learning

Orthogonality properties learning is commonly employed for feature disentanglement to enhance feature discriminability and facilitate classification [59], [60]. In the context of few-shot classification, DSN [61] generates orthogonal subspace by truncated singular value decomposition (SVD), replacing the original classifier with a set of subspace classifiers. Simultaneously, OPL [62] adopts a novel loss function to constrain the orthogonality of interclass features. POP [63] extends orthogonality learning to few-shot segmentation. It imposes constraints on the features of base and novel classes, respectively, within a two-stage framework.

However, these approaches lack the consideration of multi-dimensional orthogonal subspace, limiting the expressiveness of the representation. In contrast, our proposed method focuses on extracting discriminative object-level features in FSOD via an OSE with more dimensions. This achieves the feature disentanglement of different categories, mitigating conflicts between base and novel class representations arising from high interclass similarity in RSIs.

E. Category Imbalance

In remote sensing object detection, category imbalance is a common challenge, particularly in fine-grained datasets such as FAIR1M [64]. Prior approaches [65], [66] have attempted to alleviate this imbalance by exploiting focal loss [2] or balanced $L1$ loss [67] across all classes. In addition, from the data side, [68] solve sparse classes sample through an auto-target-duplication based on copy-paste [69]. Recently, CBCL [70] integrates contrastive learning into oriented R-CNN [71] for class imbalance, thereby enhancing the fine-grained discriminative ability of objects in RSIs.

Unlike these methods that force optimization of all classes, our BC combines base prediction guidance and equilibrium loss, effectively preventing bias toward predicting other classes.

III. Methodology

In this section, we will elucidate the pivotal components of the proposed method. We first introduce our overall isolated framework and two-stage training process. Subsequently, we provide a detailed explanation of each module, including the LoSA, OSE, and BC.

A. Overview

Our proposed BOSS framework, the BOSS detector, is shown in Fig. 3. The foundational architecture we utilize is the two-stage detector faster R-CNN. It can be substituted because our work focuses on improving fine-tuning for novel classes and avoiding the degradation of base classes, rather than altering the network's internal modules. It follows two primary phases: base pretraining with abundant data from base classes and novel fine-tuning with balanced data encompassing both base and novel classes.

In the first phase, ROI features (ROI Feats) from the proposals are projected into a learnable orthogonal subspace to

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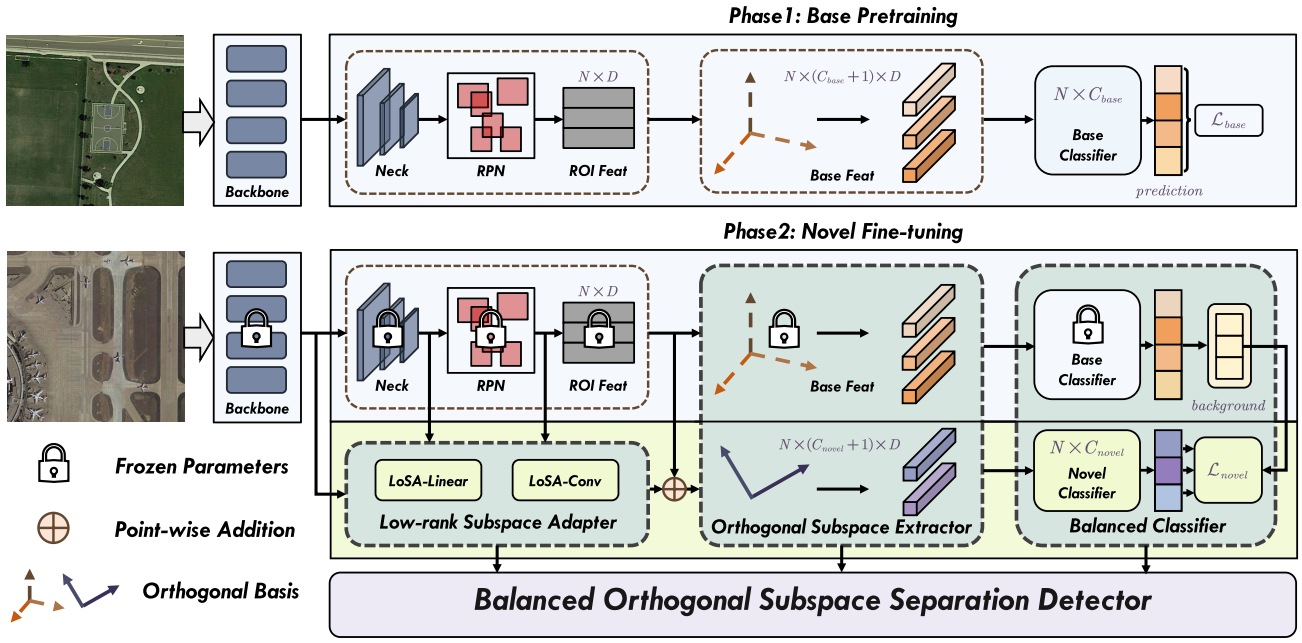


Fig. 3. Overall framework of the proposed network. In phase 1, most structures are consistent with the classic two-stage detectors, including backbone, neck, and RPN. The only difference lies in the base feat module, which indicates further extraction of discriminative base class features with an OSE. In phase 2, most of the parameters in the pretrained model will be frozen, represented by the lock in the figure.

represent disentangled characteristics of various base classes (base feat).

In the second phase, the frozen base feat, backbone, and other components exclusively contribute to detecting base classes and providing information on nonbase classes. In addition, LoSA, OSE for novel classes, and BC are introduced, given in the lower part of phase 2 in the figure. During training, the origin model interacts with these modules, transferring pretraining information to achieve generalization to novel classes. During inference, predictions are performed separately via the base and novel classifier, combined for complete detection results. Partial fine-tuning of the adapter mitigates over-fitting on few-shot novel classes while maintaining the base classes' performance. Moreover, the orthogonality between features enhances distinguishability, promoting stronger separability between base and novel classes. Finally, the BC achieves loss equilibrium for novel classes with few samples and other categories, guided by the base classifier and adjusted weights based on different predictions. Sections III-B–III-D will elaborate on these three components.

B. Low-Rank Subspace Adapter

To preserve the detection results of base classes during the fine-tuning phase, essential components from the pretraining phase, including the backbone, neck, RPN, and ROI Feat extraction are frozen. With these frozen parameters, the base classifier only outputs base class objects. For further updating the network on novel class detection, we employ a branch fine-tuning approach based on the LoSA, inspired by adapter-tuning methods in vision fundamental models and language large models. The trainable adapter is added outside the frozen structure, rather than directly updating the original model in traditional transfer-learning. Essentially, the added adapter specializes in learning to detect novel classes, achieving structural separation from the original framework. Our LoSA can modulate arbitrary convolutional and linear layers, approximating the learnable weights matrix $\widetilde{\boldsymbol{W}}$ with some low-rank matrices. Fig. 4 depicts the low-rank decomposition method we adopt, reconstructing the learnable matrix through the Hadamard product. It includes LoSA-Linear and LoSA-Conv modulation. In LoSA-Linear, the calculation process for input $\boldsymbol{x}$ is as follows:

$$\begin{aligned}\widetilde{\boldsymbol{W}} &= \boldsymbol{X_1}\boldsymbol{Y_1}^{\mathrm{T}} \odot \boldsymbol{X_2}\boldsymbol{Y_2}^{\mathrm{T}} \\ F_{\text{base}} &= \text{Linear}(\boldsymbol{W}, \boldsymbol{x}, b) \\ F_{\text{novel}} &= \text{Linear}(\widetilde{\boldsymbol{W}} + \boldsymbol{W}, \boldsymbol{x}, b) \\ \boldsymbol{y} &= \text{Cat}(F_{\text{base}}, F_{\text{novel}})\end{aligned} \tag{1}$$

where $\boldsymbol{x} \in \mathbb{R}^{N\times C_{\text{in}}}$ is the input feature, $\boldsymbol{X_1} \in \mathbb{R}^{C_{\text{out}}\times r_1}$, $\boldsymbol{X_2} \in \mathbb{R}^{C_{\text{out}}\times r_2}$, $\boldsymbol{Y_1} \in \mathbb{R}^{C_{\text{in}}\times r_1}$, and $\boldsymbol{Y_2} \in \mathbb{R}^{C_{\text{in}}\times r_2}$ are learnable parameters in a low-rank subspace compared to the original pretrained weights $\boldsymbol{W}$, and $\boldsymbol{b}$ is the pretrained bias. $\odot$ denotes Hadamard product. $F_{\text{base}}, F_{\text{novel}} \in \mathbb{R}^{N\times C_{\text{out}}}$ represent the output base and novel features. Similarly, LoSA-Conv is calculated as follows:

$$\begin{aligned}\widetilde{\boldsymbol{W}} &= \text{reshape}\left(\boldsymbol{X_1}\boldsymbol{Y_1}^{\mathrm{T}} \odot \boldsymbol{X_2}\boldsymbol{Y_2}^{\mathrm{T}}\right) \\ F_{\text{base}} &= \text{Conv2}d(\boldsymbol{W}, \boldsymbol{x}, b) \\ F_{\text{novel}} &= \text{Conv2}d\left(\widetilde{\boldsymbol{W}} + \boldsymbol{W}, \boldsymbol{x}, b\right)\end{aligned} \tag{2}$$

where $\boldsymbol{x} \in \mathbb{R}^{H\times W\times C_{\text{in}}}$, $\boldsymbol{X_1} \in \mathbb{R}^{C_{\text{out}}\times r_1}$, $\boldsymbol{X_2} \in \mathbb{R}^{C_{\text{out}}\times r_2}$, $\boldsymbol{Y_1} \in \mathbb{R}^{(C_{\text{in}}\times d\times d)\times r_1}$, $\boldsymbol{Y_2} \in \mathbb{R}^{(C_{\text{in}}\times d\times d)\times r_2}$, and $\widetilde{\boldsymbol{W}} \in \mathbb{R}^{C_{\text{out}}\times C_{\text{in}}\times d\times d}$. To meet the requirements of low-rank, the ranks of our trained weights, r_1 and r_2, are set to $r_1, r_2 \ll \min(C_{\text{in}}, C_{\text{out}})$, significantly improving optimization efficiency and memory usage. Since the pretrained over-parametrized model usually resides on a low intrinsic dimension [72], such low-rank

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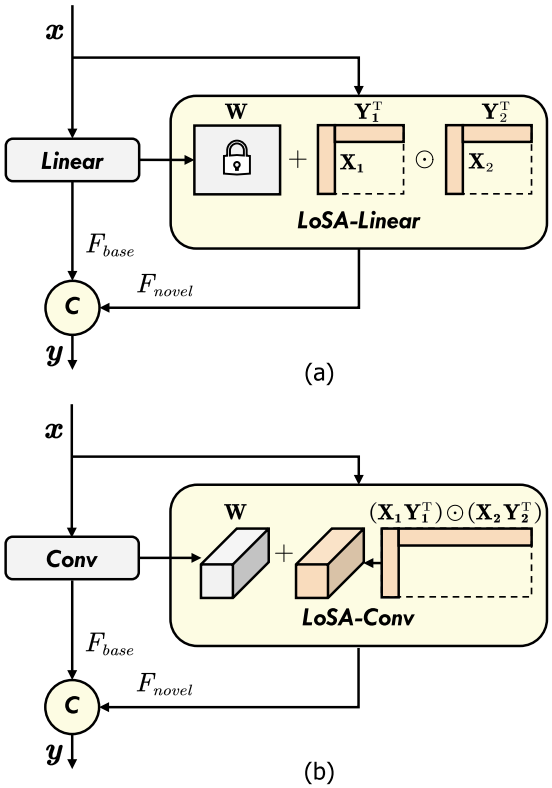


Fig. 4. Our different LoSA modulations for the convolutional and linear layers of the network. C represents a concatenate operation, which feeds the output for both base and novel classes to the next layer. (a) Linear modulation. (b) Conv modulation.

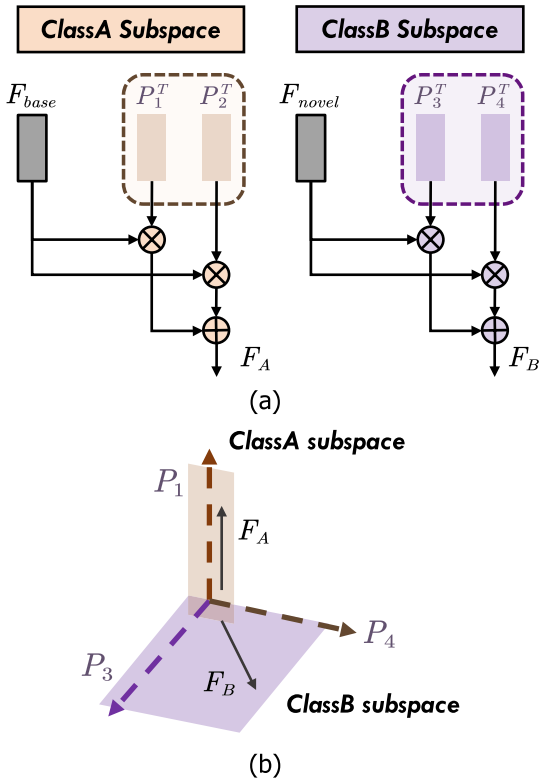


Fig. 5. Our proposed OSE. (a) Feature disentanglement of different categories within orthogonal subspace, exemplified with two classes, A and B. $\otimes$ denotes matrix multiply and $\oplus$ represents element-wise addition. (b) Relationship among multiple subspaces, resulting in the orthogonal property of features in distinct categories.

decomposition and reconstruction could be effectively trained. In contrast to the previous full-parameters fine-tuning prone to overfitting in few-shot conditions, our model alleviates this problem, attributed to the training of few parameters. In addition, to maintain the model in its original state at the start, namely, $F_{\text{base}} = F_{\text{novel}}$, $\boldsymbol{X_2}$ is zero-initialized, and other low-rank matrices are normally initialized. Subsequently, F_{novel} for each layer is gradually updated to adapt to the object detection of novel classes.

Our LoSA serves as a plug-and-play module, offering the flexibility to be optionally employed for fine-tuning either some or all modules. Consequently, adjusting input feature shapes in different modules becomes necessary. For instance, when fine-tuning only RPN and ROI Feat, candidates are generated from the feature maps output by FPN. Proposals corresponding to both base and novel branches are obtained and concatenated, resulting in twice the quantity of the pre-training phase. For $\boldsymbol{x}$ input into ROI Feat, it needs to be split as follows, yielding corresponding F_{base} and F_{novel} via LoSA:

$$\begin{aligned} \boldsymbol{x} &= [\boldsymbol{x}_{\text{base}}, \boldsymbol{x}_{\text{novel}}] \\ F_{\text{base}} &= f(\boldsymbol{W}, \boldsymbol{x}_{\text{base}}, b) \\ F_{\text{novel}} &= f(\widetilde{\boldsymbol{W}} + \boldsymbol{W}, \boldsymbol{x}_{\text{novel}}, b) \end{aligned} \tag{3}$$

where $\boldsymbol{x} \in \mathbb{R}^{2N\times C_{\text{in}}}$, $\boldsymbol{x}_{\text{base}}$, $\boldsymbol{x}_{\text{novel}} \in \mathbb{R}^{N\times C_{\text{in}}}$. f denotes linear or convolution operation.

In fact, network compression utilizes similar low-rank decomposition techniques like Tucker and CP decomposition [73], [74] to find approximations within low-rank subspace and optimize accordingly. For convenient calculation and higher dimensional reconstruction, we adopt the Hadamard product approach mentioned in (1) and (2).

In summary, our proposed LoSA achieves structural separation based on the faster R-CNN framework. It independently extracts ROI Feats for both base and novel components for subsequent steps. Notably, its flexibility facilitates the transfer of this separative design to other dense or sparse detectors.

C. Orthogonal Subspace Extractor

After obtaining ROI Feats separately from the pretrained base classes branch and the novel classes branch fine-tuned with LoSA, we introduce an OSE to disentangle features across different categories. The specific implementation is depicted in Fig. 5. For each base and novel class, we learn a set of orthogonal basis, $P_{i\times n+1}, P_{i\times n+2}, \ldots, P_{i\times n+n}$ to span a hyperplane $\mathcal{S}_i = \text{span}\{P_{i\times n+1}, P_{i\times n+2}, \ldots, P_{i\times n+n}\}, i \in \{0, 1, 2, \ldots, n_{\text{base}} + n_{\text{novel}} - 1\}$, referred to as orthogonal subspace. n is the number of bases spanning each hyperplane, n_{base} and n_{novel} represent the number of base classes and novel classes, respectively. The original features F_{base} and F_{novel} are then projected onto these hyperplanes to obtain disentangled features with dimensions $N \times (n_{\text{base}} + n_{\text{novel}}) \times C_{\text{out}}$. For simplicity, we provide an example involving only one base class A and one novel class B

$$F_A = \sum_{i=i_A+1}^{i_A+n} F_{\text{base}} P_i^{\mathrm{T}} P_i$$

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$$F_B = \sum_{j=i_B+1}^{i_B+n} F_{\text{novel}} P_j^{\mathrm{T}} P_j \tag{4}$$

where P_i and $P_j \in \mathbb{R}^{1\times C_{\text{out}}}$ are the learned parameters. i_A and i_B represent the starting index of the basis corresponding to the class A and B. F_A and $F_B \in \mathbb{R}^{N\times C_{\text{out}}}$ denote the representation of the class A and B. In the depicted case, $i_A = 0$, $i_B = 2$, $n = 2$. In fact, for enhancing the expressiveness subspace, n is set larger in practical implementation. The calculation of multiple base and novel classes is the same as F_A and F_B. This implies that the expression $F_{A^1}, F_{A^2}, \ldots, F_{A^{n_{\text{base}}}}$ will be obtained based on F_{base} for each base class $A^1, A^2, \ldots, A^{n_{\text{base}}}$ and the expression $F_{B^1}, F_{B^2}, \ldots, F_{B^{n_{\text{novel}}}}$ will be obtained based on F_{novel} for each novel class $B^1, B^2, \ldots, B^{n_{\text{novel}}}$. Furthermore, to distinguish the background, we obtain our background class representation F_{bbg} and F_{nbg} based on the above-disentangled features in the base pretraining and novel fine-tuning phases, respectively,

$$\begin{aligned} F_{bbg} &= F_{\text{base}} - \sum_{i=1}^{n_{\text{base}}} F_{A^i} \\ F_{nbg} &= F_{\text{novel}} - \sum_{j=1}^{n_{\text{novel}}} F_{B^j}. \end{aligned} \tag{5}$$

Eventually, all feature representations including background are fed into the corresponding classifier for class prediction.

In the above process, for constraining the learnable parameter P to ensure the orthogonality of the basis as follows:

$$P_i P_j^{\mathrm{T}} = \begin{cases} 0, & i \neq j \\ 1, & i = j \end{cases} \tag{6}$$

we introduce the following orthogonality losses in two phases:

$$\begin{aligned} \mathcal{L}_{\text{ortho_}pt} &= \frac{1}{N_b(N_b-1)} \sum_{i=1}^{N_b} \sum_{j>i}^{N_b} \left| P_i P_j^{\mathrm{T}} \right| \\ \mathcal{L}'_{\text{ortho_}ft} &= \frac{1}{(N_n+N_b)(N_n+N_b-1)} \sum_{i=1}^{N_n+N_b} \sum_{j>i}^{N_n+N_b} \left| P_i P_j^{\mathrm{T}} \right| \end{aligned} \tag{7}$$

where $N_b = n \cdot n_{\text{base}}$, indicating the total number of vectors to span all base classes subspace, $N_n = n \cdot n_{\text{novel}}$, representing the total number of vectors to span all novel classes subspace. It is worth noting that during novel fine-tuning, the subspace belonging to the base classes remains fixed, that is, the relevant P is frozen. With thorough optimization of (7), these vectors are constrained to be mutually orthogonal, ensuring (6). Moreover, thanks to the learned orthogonal subspace, the extracted representations are also orthogonal within their assigned hyperplanes. To illustrate, taking F_A and F_B as examples, Fig. 5(b) depicts this phenomenon. For visualization, only a part of the orthogonal basis, P_1 in the category subspace of class A is shown, and the theoretical derivation is as follows:

$$\begin{aligned} F_A F_B^{\mathrm{T}} &= \sum_{i=n_A+1}^{n_A+n} F_{\text{base}} P_i^{\mathrm{T}} P_i \sum_{j=n_B+1}^{n_B+n} P_j^{\mathrm{T}} P_j F_{\text{novel}}^{\mathrm{T}} \\ &= \sum_{i=n_A+1}^{n_A+n} \sum_{j=n_B+1}^{n_B+n} F_{\text{base}} P_i^{\mathrm{T}} \left(P_i P_j^{\mathrm{T}} \right) P_j F_{\text{novel}}^{\mathrm{T}} \\ &= 0. \end{aligned} \tag{8}$$

Therefore, our OSE enhances the independence between features, incorporating the decorrelating transformation. In addition, the proposed OSE stands out by integrating into our isolated framework BOSS. It emphases learning disentangled representation for base and novel categories, facilitating their discriminability and transfer learning to detect novel classes.

D. Balanced Classifier

To address the class imbalance in FSOD and precisely identify classes with disentangled features extracted from OSE, we design a BC module. This module consists of a base classifier and a novel classifier trained in two stages individually. In phase 2, the base classifier is entirely frozen. The training loss in phase 1 is as follows:

$$\mathcal{L}_{\text{stage1}} = \mathcal{L}_{\text{rpn}} + \mathcal{L}_{\text{reg}} + \mathcal{L}_{\text{cls}} + \mathcal{L}_{\text{ortho_}pt} \tag{9}$$

where $\mathcal{L}_{\text{cls}}$ denotes classification loss with cross-entropy loss. It is calculated based on the probability distribution $[C_b(F_{A^1}), C_b(F_{A^2}), \ldots, C_b(F_{A^{n_{\text{base}}}}), C_b(F_{bbg})]$ output by the base classifier C_b. $\mathcal{L}_{\text{reg}}$ is derived by predicting the detection bounding boxes and computing $L1$ loss, directly utilizing features F_{base} from ROI Feat. $\mathcal{L}_{\text{ortho_}pt}$ signifies orthogonality loss of the base classes subspace constructed in OSE.

During the second phase of fine-tuning, due to a limited number of novel object samples, candidate objects are labeled either as abundant background or fully trained base classes, causing a significant class imbalance. To focus on optimizing the classification loss of novel classes, the frozen base classifier is utilized to guide the sampling of the novel classifier for training. The base classifier performs classification on the extracted base class representation of N proposals, retaining only N_{sub} bounding boxes predicted as background. Subsequently, we classify novel classes based on the features of these N_{sub} bounding boxes, obtaining the predicted probability $p = [C_n(F_{B^1}), C_n(F_{B^2}), \ldots, C_n(F_{B^{n_{\text{novel}}}}), C_n(F_{nbg})]$ of novel classes according to the novel classifier C_n. In particular, when LoSA modulates the parts before ROI Feat, different candidates are generated from the base and novel branches in RPN. Accordingly, candidates from the novel branch should be input to both the subsequent base and novel branches to sample N_{sub} bounding boxes and calculate the probability distribution p. The loss is further balanced based on the distribution p, alleviating bias toward predicting base classes and background in the previous classifier caused by imbalanced class sampling in few-shot conditions. The specific balanced loss function is as follows:

$$\begin{aligned} \mathcal{L}_{\text{stage2}} &= \mathcal{L}_{\text{rpn}} + \mathcal{L}_{\text{reg}} + \mathcal{L}'_{\text{cls}} + \mathcal{L}'_{\text{ortho_}ft} \\ \mathcal{L}'_{\text{cls}} &= -\frac{1}{N_{\text{sub}} \cdot N_c} \sum_{i}^{N_{\text{sub}}} \sum_{j}^{N_c} c_{i,j} [y_{i,j} \log(p_{i,j}) \\ &\quad + (1 - y_{i,j}) \log(1 - p_{i,j})]. \end{aligned} \tag{10}$$

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In $\mathcal{L}'_{\text{cls}}$, $N_c = n_{\text{novel}} + 1$, containing the background class, $p_{i,j}$ and $y_{i,j}$ denote the predicted probability and groundtruth, indicating whether the ith detection result belongs to the jth class. $c_{i,j}$ is the weight coefficient to balance the classification loss, which is determined as follows:

$$c_{i,j} = \alpha P(j < N_c, y_{i,N_c} = 0) + \beta P(j < N_c, y_{i,N_c} = 1) + \gamma P(j = N_c, y_{i,N_c} = 0) + \theta P(j = N_c, y_{i,N_c} = 1). \tag{11}$$

When the hyperparameters α, β, γ, and θ in (11) have the same values, the losses under different distributions are equally emphasized. However, due to class imbalance, the second and fourth terms with numerous samples tend to be optimized, and novel objects are prone to be predicted as other categories. Therefore, we set α and γ to be large to emphasize the correct prediction of novel classes, reducing the penalty for prediction errors in the background class. This adjustment results in a more balanced loss, improving the model's ability to detect more novel objects. Moreover, our equilibrium loss $\mathcal{L}'_{\text{cls}}$ could be regarded as a weighted multibinary classification loss. In comparison the cross-entropy for multiclass classification, it reduces mutual suppression among different categories. Together with our completely separable structure and disentangled representation, it facilitates the independent learning of the features for each category.

Overall, our BC performs distinct classification in the base and novel classifiers. It samples appropriate novel candidates based on the guidance from the base classification results. Furthermore, during optimization, it balances loss weights across various distributions. BC effectively resolves class imbalance, prevents bias toward predicting background and base classes, and improves the detection capability of novel classes.

IV. Experiments

In this section, we first introduce two datasets for evaluating our method. Then, we explain the evaluation metrics and provide implementation details. Eventually, we conduct experiments to compare our BOSS with other methods and offer a detailed analysis of the role and overall advantages of our module.

A. Datasets

We conducted several experiments to evaluate the effectiveness of the proposed BOSS architecture on two widely used FSOD datasets in remote sensing: NWPU VHR-10 and DIOR.

1) NWPU VHR-10: The NWPU VHR-10 dataset [75] contains 800 high-resolution satellite images cropped from Google Earth and Vaihingen dataset [76]. Among them, 650 positive images contain 3775 object instances and 150 negative images only have a background. The objects are annotated in horizontal bounding box annotation format. The dataset is divided into ten categories, including airplane, ship, storage tank, baseball diamond, tennis court, basketball court, ground track field, harbor, bridge, and vehicle.

2) DIOR: DIOR [77] is a large-scale benchmark dataset for object detection in optical remote sensing images (RSIs), collecting 23 463 RSIs with image size 800×800 and 190 288 instances from Google Earth. Instances in the dataset are divided into 20 categories: airplane, airport, baseball field, basketball court, bridge, chimney, dam, expressway-service-area, expressway-toll-station, harbor, golf field, ground track field, overpass, stadium, storage tank, tennis court, train station, vehicle, windmill, and ship. The spatial resolution is 0.5–30 m.

TABLE II
Different Novel Classes Settings

Dataset	Novel classes
NWPU VHR-10	airplane, baseball diamond, tennis court
DIOR (split F [39])	airplane, baseballfield, trainstation, tenniscourt, windmill
DIOR (split 1)	baseballfield, basketballcourt, bridge, chimney, ship
DIOR (split 2)	airplane, airport, expressway-toll-station, harbor, groundtrackfield
DIOR (split 3)	dam, golffield, storagetank, tenniscourt, vehicle
DIOR (split 4)	expressway-service-area, overpass, stadium, trainstation, windmill

B. Evaluation Metric

For a fair comparison demonstrating the stable and superior performance of our work, we adopt the mAP in the PASCAL visual object classes (VOCs) challenge 2007 [78] as the evaluation metric. First, precision and recall are computed at various predicted box confidences, utilizing the intersection of union (IoU) between the predicted box and the ground truth. The IoU threshold is set to 0.5

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$
$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{12}$$

where TP, FP, and FN correspond to the number of correctly predicted samples, falsely matching the background as foreground and undetected objects; 11 confidence points from 0 to 1 with an interval of 0.1 are selected. Subsequently, based on the precisions (P) and recalls (R) for each category, the P–R curve could be plotted. The area under the P–R curve is calculated as the AP for each category

$$\text{AP}_c = \int_0^1 P_c dR. \tag{13}$$

Finally, in our experiments, three mAPs are obtained by averaging the APs of the base, novel, and total classes individually.

The precise partitioning strategy utilized by DIOR is outlined in Table II. In this context, split F aligns with the division method introduced in FSODM [39], while split 1–4 represents a completely disjoint division method commonly employed in numerous FSOD-related investigations.

C. Implementation Details

In order to demonstrate the performance of several representative algorithms and our method on the FSOD task, we first

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divided the two datasets into base and novel classes. Following the standard division, the NWPU VHR-10 dataset contains three novel classes: airplane, baseball diamond, and tennis court, and the rest are base classes. Different novel classes settings of DIOR are detailed in Table II, where split F follows FSODM [39], and split 1–4 is a completely disjoint division, with distinct novel classes. During the experiment, the base pretraining and novel fine-tuning are both performed on the trainval set, and evaluation is performed on the test set. For DIOR, it provides a public trainval set (11 725 images) and test set (11 738 images) division. However, NWPU VHR-10 has only 650 labeled positive images, where different division ratios and images are used for training and testing in various methods, lacking a unified benchmark for fair comparison. Therefore, we refer to the ratio in FSODM, allocating 2/3 of the positive images for training and 1/3 for testing. Notably, our training set is the smallest among various configurations. We then reproduce and compare different algorithms based on this data organization. We will make the divided data public to facilitate comparisons and further research within the community.

The configurations of the proposed network BOSS are detailed below. During the base pretraining on NWPU VHR-10, our model is optimized with batchsize 8 per GPU with two NVIDIA RTX3090 GPUs. It is trained for 6k, 2k and 1k iterations with learning rates of 0.02, 0.002, and $0,0002$. For the DIOR dataset, it is trained for 15k and 3k iterations with learning rates of 0.02 and 0.002. In the novel fine-tuning phase, our model is trained for 10k iterations with one NVIDIA RTX3090 GPU, adopting different learning rate scheduling strategies in different shot settings. In both stages, we apply multiscale and rotation augmentation. On the setting of our LoSA module, $r_1 = r_2 = 64$ in the LoSA-Conv, $r_1 = r_2 = 8$ in the LoSA-Linear. LoSA is utilized to modulate all the layers in FPN, RPN, and ROI Feat. In OSE, $n = 10$ and $n = 5$ correspond to base and novel classes, respectively. The hyperparameter $\alpha = 20$, $\beta = 2$, $\gamma = 10$ and $\theta = 1$ in (11) for loss equalization.

Given substantial performance variation in FSOD depending on the selected samples, we record the average results of multiple random experiments using different seeds. In each experiment, diverse images for each novel class are randomly chosen to train the model during the novel fine-tuning stage. Specifically, we train our BOSS ten times with seeds 1–10, providing representative results.

D. Experimental Results and Comparisons

1) Results on NWPU VHR-10 Dataset: To illustrate the improvement of our proposed BOSS in preserving base classes and detecting novel classes, we qualitatively compare it with the baseline model trained without separation, shown in Fig. 6. It is obvious that when training without separating base and novel classes in different stages, a large number of objects in base classes detected in pretraining will be lost, such as a vehicle, ground track field, and harbor. At the same time, the detection of novel classes is also incomplete, such as missing baseball diamonds in the second column. In contrast, BOSS ensures detection capability for base classes by preventing gradient conflicts, as shown in the fourth row of Fig. 6. Besides, due to the alleviation of class imbalance, BOSS can detect the missing baseball diamond without misidentifying it as background.

In terms of the quantitative evaluation, as shown in Table III, we compare the proposed method with one meta-learning-based method (Meta R-CNN [34]) and three transfer-learning-based methods (TFA [21], FSCE [20], and CIR-FSD [45]) in the configuration of three-shot, five-shot, and ten-shot. The marked † indicates that the results are averaged over multiple runs. It can be seen that with an increase in the number of samples, the base mAP in Two-Stage BOSS could be stably comparable to the pretrained model, One-Stage BOSS, exhibiting only marginal decreases of 0.9%, 1.3%, and 1.2%, respectively. In comparison to the classical transfer learning method TFA, the performance in base classes improves by about 26.8%, 28.5%, and 25.7%. For novel categories, because our BOSS avoids the gradient conflict and suppresses the adverse effects of class imbalance, it brings improvements of 12.4%, 19.5%, and 24.7%, which increases with the number of shots. Despite Meta R-CNN based on meta-learning or recent advancements CIR-FSD based on transfer-learning does not have a significant drop in base classes performance, they are still inferior to our method, especially for novel classes. In addition, our BOSS refrains from excessive adjustments to the internal network structure, emphasizing the construction of a completely separated framework. This approach is orthogonal to most methods of modifying the convolution module or deepening the network layers, supporting further expansion of our model. In summary, the results of our work on the NWPU VHR-10 dataset outperform classic and advanced algorithms. A more detailed comparison with other methods will be provided in Section IV-D2 using the more intricate DIOR dataset.

2) Results on DIOR Dataset: On the DIOR dataset, we conduct five-shot, ten-shot, and 20-shot experiments under five base/novel classes splits, and the detection results are visualized under split F and split 1 in Fig. 7. In fact, in the more complex scenarios of the DIOR dataset, our approach exhibits more pronounced advantages. For instance, in the first column of split 1, suffering from the optimization conflict between the base and novel classes, the baseline method mistakenly predicts a base class object (train station) as a novel class (airport) during the novel fine-tuning stage. On the contrary, our proposed BOSS could retain accurate detection results of base classes, thanks to the independent learning of base and novel classes. Furthermore, updating the regression branch in the fine-tuning phase also enhances the base class detection. As a result, it yields superior performance compared to the pretrained model, effectively suppressing instances of false positives, as displayed in the second column of split F.

In terms of the quantitative evaluation, we compare the proposed method with three meta-learning based methods (Meta R-CNN [34], FSODM [39], and TINet [42]) and four transfer-learning based methods (TFA [21], FSCE [20], CIR-FSD [45], and PAMS-Det [44]) in split F, and compare with three meta-learning based methods (Meta R-CNN [34], P-CNN [43], and MM R-CNN [41]) and three transfer-learning

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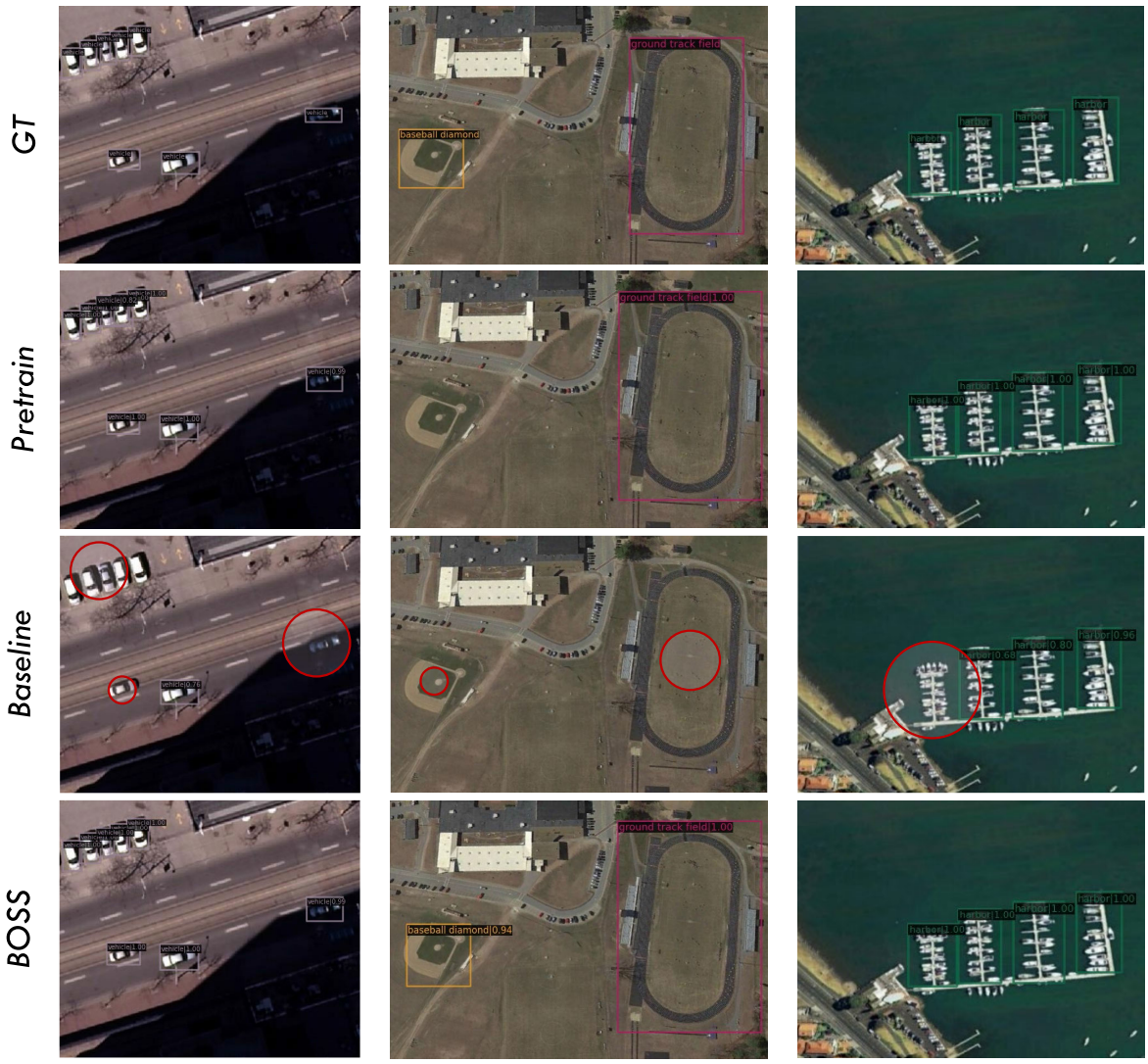


Fig. 6. Visualization of detection results of NWPU VHR-10 dataset. Pretraining and BOSS both come from the results of our BOSS model, and the baseline is the model fine-tuned by the TFA method. Only the bounding boxes with a confidence greater than 0.5 are visualized, and the model setting is ten-shot. The area circled in red indicates the incorrect or missing predictions in the baseline.

TABLE III

FSOD PERFORMANCE OBTAINED BY DIFFERENT METHODS FOR THE NWPU VHR-10 DATASET (OPTIMAL RESULTS ARE BOLDED)

Method	3-shot			5-shot			10-shot		
	Base	Novel	Total	Base	Novel	Total	Base	Novel	Total
Meta R-CNN†	0.813	0.045	0.583	0.804	0.046	0.577	0.823	0.061	0.595
TFA†	0.594	0.229	0.484	0.573	0.293	0.489	0.602	0.307	0.514
FSCE†	0.702	0.270	0.572	0.713	0.404	0.620	0.733	0.471	0.655
CIR-FSD†	0.831	0.302	0.672	0.839	0.310	0.700	0.839	0.422	0.731
One-Stage BOSS (ours)	0.871	-	-	0.871	-	-	0.871	-	-
Two-Stage BOSS†(ours)	**0.862**	**0.353**	**0.709**	**0.858**	**0.488**	**0.747**	**0.859**	**0.554**	**0.767**

based methods (TFA [21], FSCE [20], and FsCIT [49]) in split 1–4, as shown in Tables IV and V. The results without † are the best of multiple runs, and those with † are averaged. It can be seen that our model achieves the best results in various experimental conditions regardless of base or novel classes. Specifically, in split F, our method on average outperforms the state-of-the-art method TINet by 4.7%, 0.6%, and 1.5%, respectively, on the novel classes, even compared to its single run. Although the performance of our method in the five-shot setting is close to that of CIR-FSD, only 0.3% higher, it exhibits notable advantages in ten-shot and 20-shot, higher than CIR-FSD by 1.0% and 1.7%, respectively. Furthermore, our optimal performance of multiple runs significantly exceeds other methods. In other division strategies split 1–4, our method is much higher than the advanced methods FsCIT and MM-RCNN, especially the improvement on the base classes is around 15%–25%, which is stably better than one-stage BOSS, proving that the proposed BOSS competently maintains the performance of the pretrained model on base classes and promotes knowledge transfer to novel classes. Regarding

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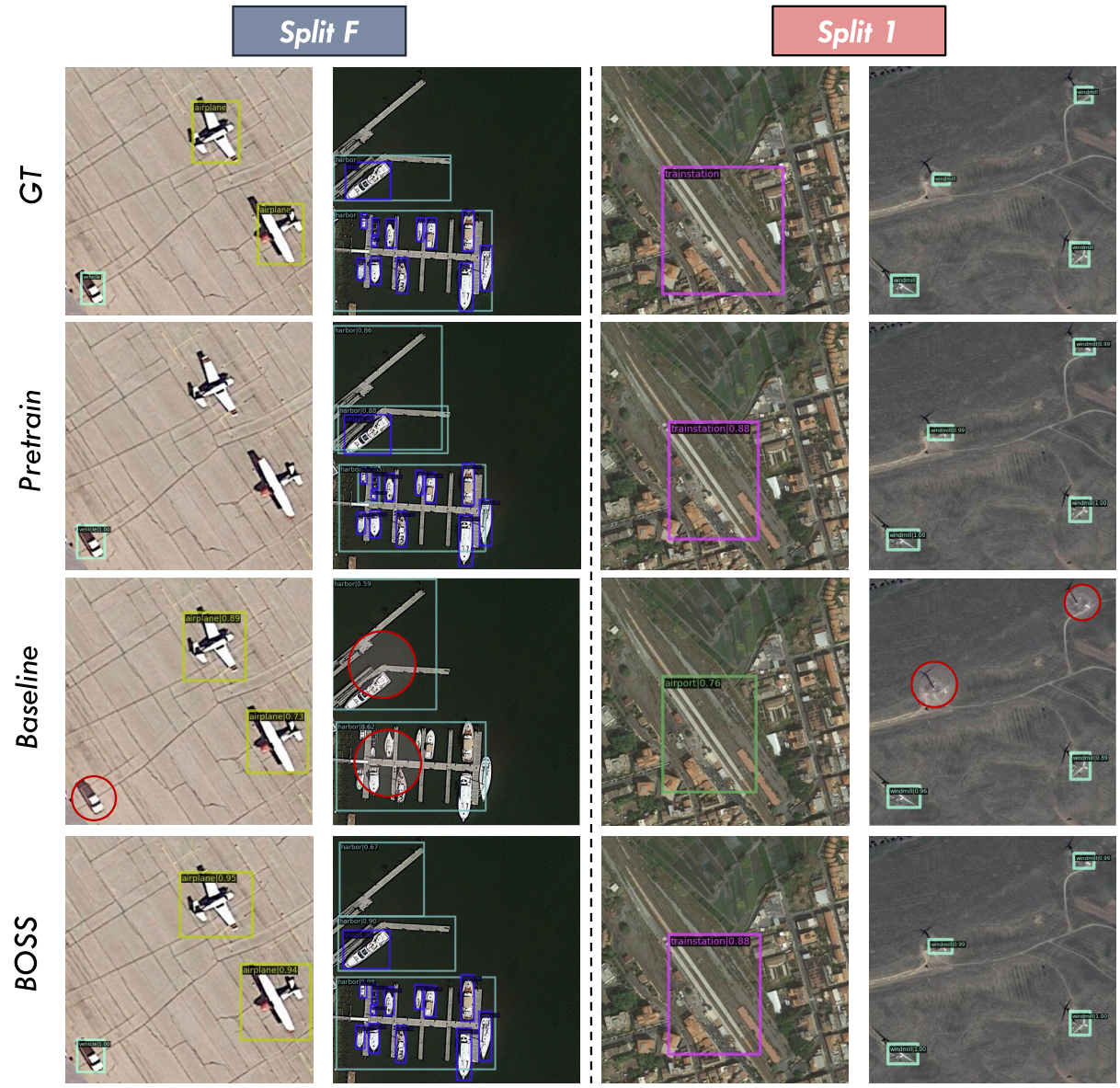


Fig. 7. Visualization of detection results of DIOR dataset. Only the bounding boxes with a confidence greater than 0.5 are visualized, and the model setting is ten-shot. The area circled in red indicates the incorrect or missing prediction in the base pretraining and novel baseline.

TABLE IV

FSOD PERFORMANCE OBTAINED BY DIFFERENT METHODS FOR THE DIOR UNDER SPLIT *F* (OPTIMAL RESULTS ARE BOLDED)

split config	Method	5-shot			10-shot			20-shot		
		Base	Novel	Total	Base	Novel	Total	Base	Novel	Total
Split F	Meta R-CNN†	-	0.154	-	-	0.199	-	-	-	-
	TFA†	0.518	0.226	0.445	0.604	0.255	0.517	0.535	0.375	0.495
	FSCE†	-	0.282	-	-	0.346	-	-	-	-
	CIR-FSD†	-	0.330	-	-	0.380	-	-	0.430	-
	FSODM	0.535	0.246	0.463	0.535	0.320	0.482	0.535	0.366	0.493
	PAMS-Det	0.651	0.282	0.559	0.651	0.328	0.570	0.651	0.384	-
	TINet	0.578	0.286	0.505	0.574	0.384	0.527	0.621	0.432	0.574
	One-Stage BOSS (ours)	0.733	-	-	0.733	-	-	0.733	-	-
	Two-Stage BOSS†(ours)	**0.745**	**0.333**	**0.642**	**0.748**	**0.390**	**0.659**	**0.753**	**0.447**	**0.676**
	Two-Stage BOSS(ours)	**0.744**	**0.370**	**0.650**	**0.754**	**0.409**	**0.668**	**0.753**	**0.478**	**0.684**

the novel mAP, apart from split 3 which is comparable to MM R-CNN, our other results show obvious improvements. Especially for split 1, the average results of BOSS improve by 13.0%, 13.1%, and 16.1% compared with MM R-CNN in the three few-shot settings. Compared with FsCIT in ten-shot and 20-shot, the best results of BOSS improve 17.3% and 14.2%. Overall, the optimal performance of BOSS across various configurations of DIOR demonstrates that it effectively mitigates

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TABLE V
FSOD PERFORMANCE OBTAINED BY DIFFERENT METHODS FOR THE DIOR UNDER SPLITS 1–4 (OPTIMAL RESULTS ARE BOLDED)

split config	Method	5-shot			10-shot			20-shot		
		Base	Novel	Total	Base	Novel	Total	Base	Novel	Total
Split 1	Meta R-CNN†	0.482	0.168	-	0.494	0.220	-	0.493	0.243	-
	TFA†	0.488	0.162	-	0.492	0.229	-	0.491	0.237	-
	FSCE†	0.496	0.178	-	0.504	0.239	-	0.509	0.245	-
	P-CNN†	0.484	0.228	-	0.509	0.276	-	0.522	0.296	-
	MM R-CNN†	0.488	0.239	-	0.501	0.288	-	0.529	0.308	-
	FsCIT	-	-	-	0.602	0.281	0.522	0.640	0.340	0.565
	One-Stage BOSS (ours)	0.753	-	-	0.753	-	-	0.753	-	-
	Two-Stage BOSS†(ours)	**0.756**	**0.369**	**0.659**	**0.756**	**0.419**	**0.672**	**0.756**	**0.469**	**0.684**
	Two-Stage BOSS(ours)	**0.757**	**0.394**	**0.666**	**0.755**	**0.454**	**0.680**	**0.756**	**0.482**	**0.688**
Split 2	Meta R-CNN†	0.502	0.125	-	0.503	0.153	-	0.523	0.205	-
	TFA†	0.495	0.036	-	0.485	0.052	-	0.485	0.078	-
	FSCE†	0.498	0.086	-	0.506	0.084	-	0.493	0.089	-
	P-CNN†	0.491	0.149	-	0.525	0.189	-	0.516	0.228	-
	MM R-CNN†	0.499	0.155	-	0.521	0.201	-	0.525	0.239	-
	One-Stage BOSS (ours)	0.743	-	-	0.743	-	-	0.743	-	-
	Two-Stage BOSS†(ours)	**0.753**	**0.210**	**0.617**	**0.753**	**0.261**	**0.630**	**0.752**	**0.343**	**0.650**
Split 3	Meta R-CNN†	0.500	0.154	-	0.522	0.203	-	0.537	0.263	-
	TFA†	0.524	0.138	-	0.528	0.146	-	0.524	0.189	-
	FSCE†	0.544	0.149	-	0.517	0.171	-	0.541	0.204	-
	P-CNN†	0.499	0.188	-	0.521	0.234	-	0.531	0.288	-
	MM R-CNN†	0.495	0.197	-	0.524	0.250	-	0.536	0.305	-
	One-Stage BOSS (ours)	0.756	-	-	0.756	-	-	0.756	-	-
	Two-Stage BOSS†(ours)	**0.760**	**0.206**	**0.621**	**0.758**	**0.251**	**0.631**	**0.765**	**0.35**	**0.661**
Split 4	Meta R-CNN†	0.512	0.136	-	0.523	0.170	-	0.534	0.220	-
	TFA†	0.496	0.096	-	0.485	0.114	-	0.502	0.128	-
	FSCE†	0.508	0.111	-	0.497	0.136	-	0.515	0.142	-
	P-CNN†	0.499	0.175	-	0.517	0.189	-	0.523	0.257	-
	MM R-CNN†	0.507	0.187	-	0.521	0.203	-	0.532	0.271	-
	One-Stage BOSS (ours)	0.732	-	-	0.732	-	-	0.732	-	-
	Two-Stage BOSS†(ours)	**0.745**	**0.195**	**0.608**	**0.749**	**0.231**	**0.620**	**0.749**	**0.337**	**0.646**

TABLE VI
ABLATION OF DIFFERENT COMPONENTS

LoSA	OSE	BC	Base	Novel	Total
×	×	×	0.582	0.282	0.507
✓	×	×	0.714	0.328	0.618
✓	✓	×	0.745	0.391	0.657
✓	✓	✓	**0.745**	**0.420**	**0.664**

TABLE VII
ABLATION OF DIFFERENT MODULES MODULATED BY LOSA

FPN	RPN	ROI Feat	Base	Novel	Total	FLOPs
×	×	LoSA	0.742	0.262	0.622	302.5G
×	Full-Param	LoSA	0.737	0.389	0.650	302.5G
×	LoSA	LoSA	0.745	0.381	0.654	450.6G
LoSA	Full-Param	LoSA	0.738	0.407	0.655	487.4G
LoSA	LoSA	LoSA	**0.745**	**0.420**	**0.664**	487.4G

optimization conflicts between base and novel classes and addresses class imbalance, thereby enhancing detection in all categories. Our separation detector proves adaptable to FSOD in challenging remote sensing scenarios for diverse categories.

E. Ablation Studies

In the ablation studies, we explore the contribution of each proposed module to FSOD by removing them or adjusting their internal parameters in the full BOSS architecture. All experiments are performed based on the ten-shot setting in split *F* of the DIOR dataset, with a fixed random seed 1. The specific experimental analysis is as follows.

1) Ablation Study of Different Components: For investigating the individual contribution of the proposed three modules, LOSA, OSE, and BC in the BOSS framework, we perform an ablation study on different components, as detailed in Table VI.

It is evident that in the absence of these three modules, the network reverts to the original TFA, only with 58.2% base mAP, 28.2% novel mAP, and 50.7% total mAP. Upon integrating LoSA, there is a respective increase of 13.2% and 4.6% for base and novel classes. The significant enhancement in base class performance indicates that our structural separation mitigates gradient conflict, preventing base class degradation to a certain extent. The introduction of OSE further elevates the base mAP by 3.1% and the novel mAP by 6.3%. Under the joint influence of LOSA and OSE, gradient conflict is completely avoided, resulting in improved base class performance that even exceeds the pretrained model. In addition, the enhanced separability from feature distanglement of different classes contributes to substantial improvements in novel class detection. Eventually, the incorporation of BC addresses class imbalance, leading to a 2.9% improvement in novel mAP. The base classes are unaffected.

In conclusion, each of our three modules plays a pivotal role in addressing FSOD challenges in RSIs. They retain the model's capability to detect base classes and facilitate

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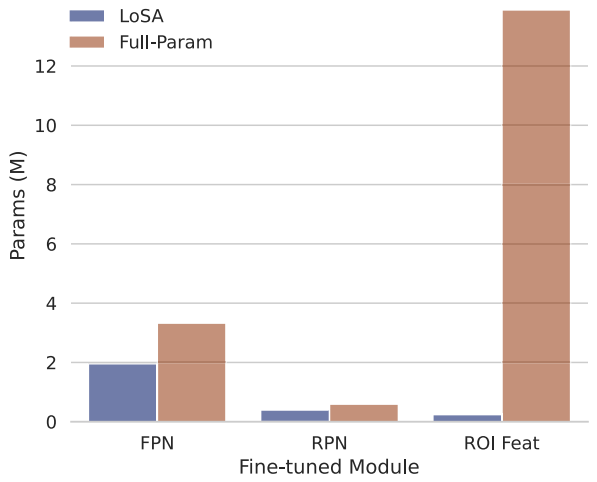


Fig. 8. Trainable params for different modules. All params are calculated as float type.

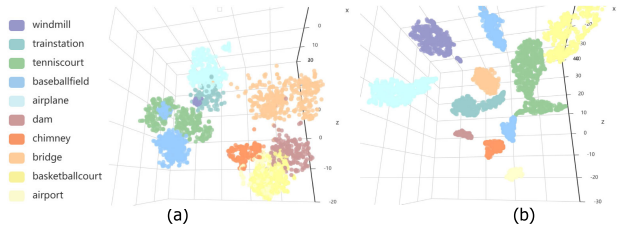


Fig. 9. Visualization of disentangled features. We project the original 1024-D features into a 3-D space using t-SNE [79] for clarity in observation. We select five base classes (airport, basketball court, bridge, chimney, and dam) and five novel classes (airplane, baseball field, tennis court, train station, and windmill). (a) Baseline. (b) BOSS.

the learning of novel classes, collectively contributing to our BOSS architecture and achieving state-of-the-art performance.

2) Modulation of LoSA Module: To analyze the effects of LoSA modulating different components, we compare the performance of different fine-tuning strategies in Table VII, where Full-Param represents fine-tuning and replacing all original parameters.

Initially, we exclusively fine-tune ROI Feat using LoSA, reaching 74.2% base mAP. This indicates that even partial structural separation can effectively prevent the degradation of base classes caused by gradient conflict. However, due to the limited fine-tuning parameters, the model struggles to adapt to novel classes, with a notably low novel mAP of 26.2%. Subsequent full fine-tuning of RPN significantly enhances the novel mAP by 12.7%, with a minor impact on base classes, which experience a decrease of 0.5%. Conversely, LoSA fine-tuning of RPN boosts up the novel mAP from 26.2% to 38.1%, accompanied by a slight enhancement of 0.3% base mAP. This reflects that more fine-tuned modules aid in fully optimizing the network for novel classes, and the LoSA fine-tuning method ensures that unfreezing new modules does not compromise the base class again. After fine-tuning FPN, the novel mAP achieves a further improvement of 1.8% and 3.9% under two fine-tuning methods of LoSA RPN and Full-Param RPN, respectively. The base mAP remains relatively stable without the degradation problem. It is apparent that when all the modules are fine-tuned via LoSA modulation, the best model is obtained. Taken together, these results suggest that our LoSA module supports flexible parameter updating of more components for few-shot data, confirming its effectiveness.

To illustrate the cost of our LoSA in various components for fine-tuning, we compare FLOPs and the training parameters between full fine-tuning and LoSA modulation across various modules as depicted in Table VII and Fig. 8. For FLOPs, although LoSA in RPN introduces an extra computational overhead of approximately 49% compared to Full-Param which does not increase GLFOPs, this is imperative for fine-tuning all components. When FPN is further modulated with LoSA, it only brings 8.2% computational cost, with the added advantage that the LoSA in RPN yields performance improvements without redundant computations. With the significant performance gain, we believe that such computational cost is worthwhile and the efficiency can also be assured. For Params, the analysis reveals that the introduced LoSA demands fewer training parameters than the original layer within each module. In particular, the parameters associated with ROI Feat exhibit a significant reduction of 95%. Consequently, LoSA not only improves the optimization capacity of the network but also promotes adaptability to diverse architectures, owing to its lightweight characteristics.

3) Expressiveness of OSE Module: By visualizing the disentangled features obtained from both the baseline and BOSS model in Fig. 9, we elucidate that the proposed OSE module facilitates the separability of features from distinct categories. The figure illustrates the feature projection distribution of each object onto the learned orthogonal subspace corresponding to its ground truth. In the baseline model, intraclass features exhibit greater dispersion, leading to notable overlap in the feature space within various classes. This overlap is evident not only among base classes (such as bridges, airports, and dams) but also between base and novel classes (bridges and train stations), posing difficulties in distinguishing features. Thanks to our OSE module, the clusters of features for each category obtained by BOSS have no overlap. Consequently, it becomes feasible to learn classification hyperplanes for distinguishing different categories using linear classifiers. It proves that our feature disentanglement based on orthogonal subspace can maximize the discriminability of classes in RSIs with high interclass similarity. Even with few shots, the learned novel features retain sufficient expressiveness in space.

In order to further explore the effects of enhanced feature expressiveness resulting from increased subspace dimensions on FSOD performance, we conduct the ablation experiments on subspace dimensions n of base and novel classes. The results of the novel mAP are presented in Fig. 10.

In the absence of orthogonality constraints within the subspace, we observe the lowest novel mAP of 37.4%, indicating insufficient separability among features. Constraining 1-D vectors spanning all the subspace to be mutually orthogonal results in a 1.8% increase in mAP, highlighting the efficacy of orthogonality learning in enhancing feature distinguishability. As the subspace dimension grows, the learned features are not limited to a low-dimensional subset, leading to further

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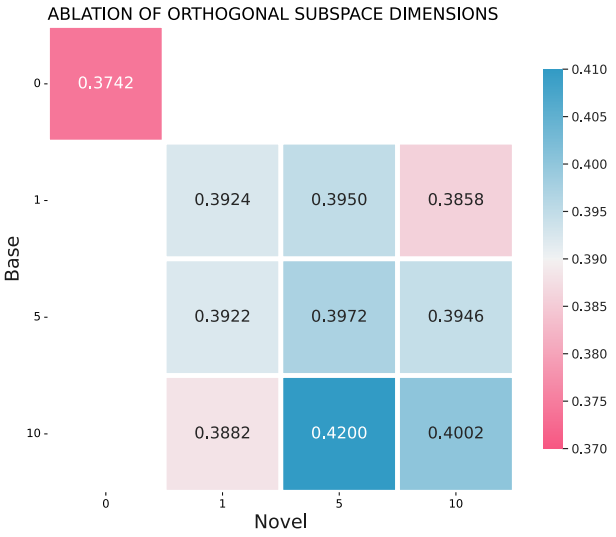


Fig. 10. Ablation of orthogonal subspace dimensions. Both the base and novel classes adopt the setting of $n = 0$, 1, 5, and 10, where 0 means that only a 1-D vector is used to expand the subspace without orthogonality constraint.

performance improvement. Notably, setting n of base and novel subspace to 10 and 5, respectively, the model achieves the optimal performance of 42.0%. However, setting n of the novel subspace to 10 leads to a performance decrease, irrespective of n of the base subspace. This is attributed to the fact that higher novel subspace dimensions bring representation redundancy, especially fitting few-shot data, which potentially interferes with the effective learning of features. The above results validate that our orthogonal feature extraction is beneficial for classification. As the feature expressiveness increases within a specific range, they tend to become more discriminative.

4) Hyperparameters of BC Module: Moreover, to analyze the impact of classifier weight balancing on FSOD, we perform experiments with varying hyperparameters α and γ in the BC module. These hyperparameters control the loss of weight of novel class prediction.

In Table VIII, a low value of α ($\alpha = 1$) severely impairs the detection performance of novel classes, with only 35.0% mAP. This is attributed to evident class imbalance, where excessive background sampling hinders foreground detection, resulting in minimal emphasis on novel class optimization. Subsequently, increasing α to 5 leads to a significant improvement by 5.7% in novel mAP, while the performance of the base class remains unaffected. This is achieved by reinforcing the penalty on novel classes, mitigating the issue of the network excessively focusing on background sampling and missing novel objects. Further increasing α to 20, our model obtains the optimal result of 42.0%, showing the significance of balanced weight configuration. However, as α increases to 30, the heightened emphasis on novel objects causes more false positives compared to the increase in correct detections, and novel mAP drops by 1.3%. Consequently, we choose $\alpha = 20$ as the final adopted parameter.

Regarding the hyperparameter γ in Table IX, it exhibits lower sensitivity in comparison. Even when γ takes a small

TABLE VIII
ABLATION OF HYPERPARAMETERS α

	$\alpha = 1$	$\alpha = 5$	$\alpha = 10$	$\alpha = 20$	$\alpha = 30$
Base	0.745	0.745	0.745	**0.745**	0.745
Novel	0.350	0.407	0.408	**0.420**	0.407
Total	0.643	0.658	0.657	**0.664**	0.657

TABLE IX
ABLATION OF HYPERPARAMETERS γ

	$\gamma = 1$	$\gamma = 5$	$\gamma = 10$	$\gamma = 15$	$\gamma = 25$
Base	0.745	0.745	**0.745**	0.745	0.745
Novel	0.401	0.408	**0.420**	0.404	0.400
Total	0.656	0.657	**0.664**	0.657	0.655

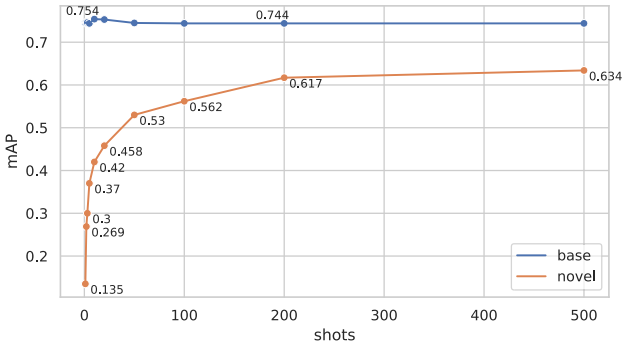


Fig. 11. Ablation study for a number of shots. The number of shots selected is 1, 2, 3, 5, 10, 20, 50, 100, 200, and 500. For the novel classes, mAPs of all shot configurations are given, while for the base classes, only the highest and lowest are given.

value, such as 1, the mAP is only 1.9% lower than the best result when $\gamma = 10$. Similarly, further increasing γ beyond 10 leads to a decrease in mAP, so we adopt of $\gamma = 10$ as the default setting. The performance of base classes remains consistent with that of Table VIII, which does not vary with the weight. This observation also reflects that the gradients conflict which makes the base performance degraded, has been effectively resolved by LoSA and OSE. In summary, to fulfill loss equilibrium in our BC, it is essential to impose sufficient penalties on mispredictions of novel categories.

5) Number of Shots: We analyze the relationship between the number of novel shots, which is used in the fine-tuning phase, and the model performance, as illustrated in Fig. 11. As the number of samples increases, the base mAP remains relatively stable, with minor fluctuations ranging from 74.4% to 75.4%. In contrast, the novel class exhibits substantial gains, particularly with fewer shots, showing more pronounced advancements. For instance, increasing from one-shot to two-shot results in a boost of about 13.4% with just one additional sample. This improvement trend gradually diminishes as the number of samples becomes larger, eventually converging around 200-shot, with an approximate increase of merely 1.7% when extended to 500 shots. The results not only highlight the stable performance of our completely separated framework in preventing base class degeneration but also demonstrate its capacity for generalization across novel classes.

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V. Conclusion

In conclusion, this article proposes an overall isolated two-stage detection architecture named BOSS detector for FSOD. The key contributions can be summarized as follows. The LoSA enables structural separation for the training of base and novel classes by modulating different parts of the network. The OSE learns disentangled feature representation, enhancing the separability of each base and novel class for more adaptive updates in two branches. The BC addresses class-imbalanced sampling through guidance from the base classifier and weight assignment in the novel classifier based on different result distributions. These modules are demonstrated to support the preservation of base classes and the learning of novel classes in the conducted experiments.

The findings of this work highlight the significance of structure separation, feature disentanglement, and loss equilibrium, thoroughly solving optimization contradiction and class imbalance in remote sensing scenarios. Future work will delve into exploring more application tasks, such as few-shot fine-grained and open-vocabulary object detection, based on our proposed BOSS framework.

References

[1] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 6, pp. 1137–1149, Jun. 2017.

[2] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2999–3007.

[3] Z. Yang, S. Liu, H. Hu, L. Wang, and S. Lin, "RepPoints: Point set representation for object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 9656–9665.

[4] Z. Tian, C. Shen, H. Chen, and T. He, "FCOS: Fully convolutional one-stage object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 9626–9635.

[5] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, and S. Zagoruyko, "End-to-end object detection with transformers," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2020, pp. 213–229.

[6] S. Zhang, C. Chi, Y. Yao, Z. Lei, and S. Z. Li, "Bridging the gap between anchor-based and anchor-free detection via adaptive training sample selection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 9756–9765.

[7] J. Ding, N. Xue, Y. Long, G.-S. Xia, and Q. Lu, "Learning RoI transformer for oriented object detection in aerial images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 2849–2858.

[8] Y. Xu et al., "Gliding vertex on the horizontal bounding box for multi-oriented object detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 4, pp. 1452–1459, Apr. 2021.

[9] X. Yang, J. Yan, Z. Feng, and T. He, "R^{3}Det: Refined single-stage detector with feature refinement for rotating object," in *Proc. AAAI Conf. Artif. Intell.*, 2021, pp. 3163–3171.

[10] J. Han, J. Ding, N. Xue, and G. Xia, "ReDet: A rotation-equivariant detector for aerial object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 2786–2795.

[11] J. Han, J. Ding, J. Li, and G.-S. Xia, "Align deep features for oriented object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5602511.

[12] X. Yang et al., "Learning high-precision bounding box for rotated object detection via Kullback–Leibler divergence," in *Proc. Adv. Neural Inf. Process. Syst.*, M. Ranzato, A. Beygelzimer, Y. N. Dauphin, P. Liang, and J. W. Vaughan, Eds., 2021, pp. 18381–18394.

[13] X. Yang, J. Yan, Q. Ming, W. Wang, X. Zhang, and Q. Tian, "Rethinking rotated object detection with Gaussian Wasserstein distance loss," in *Proc. 38th Int. Conf. Mach. Learn. (ICML)*, in Proceedings of Machine Learning Research, vol. 139, M. Meila and T. Zhang, Eds., Jul. 2021, pp. 11830–11841.

[14] Z. Guo, C. Liu, X. Zhang, J. Jiao, X. Ji, and Q. Ye, "Beyond bounding-box: Convex-hull feature adaptation for oriented and densely packed object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 8792–8801.

[15] X. Wu, D. Sahoo, and S. Hoi, "Meta-RCNN: Meta learning for few-shot object detection," in *Proc. 28th ACM Int. Conf. Multimedia*, Oct. 2020, pp. 1679–1687.

[16] G. Zhang et al., "Meta-DETR: Image-level few-shot detection with inter-class correlation exploitation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 11, pp. 12832–12843, Nov. 2023.

[17] L. Qiao et al., "DeFRCN: Decoupled faster R-CNN for few-shot object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 8661–8670.

[18] G. Han, J. Ma, S. Huang, L. Chen, and S. Chang, "Few-shot object detection with fully cross-transformer," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 5311–5320.

[19] B. Gao et al., "Decoupling classifier for boosting few-shot object detection and instance segmentation," in *Proc. Adv. Neural Inf. Process. Syst.*, 2022, pp. 18640–18652.

[20] B. Sun, B. Li, S. Cai, Y. Yuan, and C. Zhang, "FSCE: Few-shot object detection via contrastive proposal encoding," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 7352–7362.

[21] X. Wang, T. E. Huang, T. Darrell, J. E. Gonzalez, and F. Yu, "Frustratingly simple few-shot object detection," in *Proc. Int. Conf. Mach. Learn.*, 2020, pp. 9919–9928.

[22] Z. Fan, Y. Ma, Z. Li, and J. Sun, "Generalized few-shot object detection without forgetting," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 4527–4536.

[23] J. Ma, Y. Niu, J. Xu, S. Huang, G. Han, and S.-F. Chang, "DiGeo: Discriminative geometry-aware learning for generalized few-shot object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 3208–3218.

[24] P. Sun et al., "Sparse R-CNN: End-to-end object detection with learnable proposals," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 14454–14463.

[25] X. Zhu, W. Su, L. Lu, B. Li, X. Wang, and J. Dai, "Deformable DETR: Deformable transformers for end-to-end object detection," in *Proc. Int. Conf. Learn. Represent.*, 2021, pp. 1–16.

[26] D. Meng et al., "Conditional DETR for fast training convergence," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 3631–3640.

[27] S. Liu et al., "DAB-DETR: Dynamic anchor boxes are better queries for DETR," in *Proc. Int. Conf. Learn. Represent.*, 2022, pp. 1–20.

[28] H. Zhang et al., "DINO: DETR with improved denoising anchor boxes for end-to-end object detection," in *Proc. Int. Conf. Learn. Represent.*, 2023, pp. 1–19.

[29] H. Hu, J. Gu, Z. Zhang, J. Dai, and Y. Wei, "Relation networks for object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 3588–3597.

[30] K. He, G. Gkioxari, P. Dollár, and R. Girshick, "Mask R-CNN," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 42, no. 2, pp. 386–397, Feb. 2020.

[31] Z. Cai and N. Vasconcelos, "Cascade R-CNN: Delving into high quality object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 6154–6162.

[32] W. Li, Y. Chen, K. Hu, and J. Zhu, "Oriented RepPoints for aerial object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 1819–1828.

[33] Y. Zeng, X. Yang, Q. Li, Y. Chen, and J. Yan, "ARS-DETR: Aspect ratio sensitive oriented object detection with transformer," 2023, *arXiv:2303.04989*.

[34] X. Yan, Z. Chen, A. Xu, X. Wang, X. Liang, and L. Lin, "Meta R-CNN: Towards general solver for instance-level low-shot learning," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 9576–9585.

[35] H. Chen, Y. Wang, G. Wang, and Y. Qiao, "LSTD: A low-shot transfer detector for object detection," in *Proc. 30th AAAI Conf. Artif. Intell.*, Feb. 2018, pp. 2836–2843.

[36] C. Baur, S. Albarqouni, and N. Navab, "Semi-supervised deep learning for fully convolutional networks," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assisted Intervent.*, vol. 10435, M. Descoteaux, L. Maier-Hein, A. M. Franz, P. Jannin, D. L. Collins, and S. Duchesne, Eds., 2017, pp. 311–319.

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[37] E. Protopapadakis, A. Doulamis, N. Doulamis, and E. Maltezos, "Stacked autoencoders driven by semi-supervised learning for building extraction from near infrared remote sensing imagery," *Remote Sens.*, vol. 13, no. 3, p. 371, Jan. 2021.

[38] Y. Ouali, C. Hudelot, and M. Tami, "An overview of deep semi-supervised learning," 2020, *arXiv:2006.05278*.

[39] X. Li, J. Deng, and Y. Fang, "Few-shot object detection on remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5601614.

[40] Y. Zhang, B. Zhang, and B. Wang, "Few-shot object detection with self-adaptive global similarity and two-way foreground stimulator in remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 7263–7276, 2022.

[41] J. Li et al., "MM-RCNN: Toward few-shot object detection in remote sensing images with meta memory," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5635114.

[42] N. Liu, X. Xu, T. Celik, Z. Gan, and H.-C. Li, "Transformation-invariant network for few-shot object detection in remote-sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5625314.

[43] G. Cheng et al., "Prototype-CNN for few-shot object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5604610.

[44] Z. Zhao, P. Tang, L. Zhao, and Z. Zhang, "Few-shot object detection of remote sensing images via two-stage fine-tuning," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[45] Y. Wang, C. Xu, C. Liu, and Z. Li, "Context information refinement for few-shot object detection in remote sensing images," *Remote Sens.*, vol. 14, no. 14, p. 3255, Jul. 2022.

[46] Y. Zhou, H. Hu, J. Zhao, H. Zhu, R. Yao, and W.-L. Du, "Few-shot object detection via context-aware aggregation for remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[47] S. Ma et al., "Automatic aug-aware contrastive proposal encoding for few-shot object detection of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5615211.

[48] X. Lu et al., "Few-shot object detection in aerial imagery guided by text-modal knowledge," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5604719.

[49] L. Li, X. Yao, X. Wang, D. Hong, G. Cheng, and J. Han, "Robust few-shot aerial image object detection via unbiased proposals filtration," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5617011.

[50] A. Radford et al., "Learning transferable visual models from natural language supervision," in *Proc. Int. Conf. Mach. Learn.*, vol. 139, 2021, pp. 8748–8763.

[51] N. Houlsby et al., "Parameter-efficient transfer learning for NLP," in *Proc. Int. Conf. Mach. Learn.*, 2019, pp. 2790–2799.

[52] J. Pfeiffer, A. Kamath, A. Rücklé, K. Cho, and I. Gurevych, "AdapterFusion: Non-destructive task composition for transfer learning," in *Proc. 16th Conf. Eur. Chapter Assoc. Comput. Linguistics*, 2021, pp. 487–503.

[53] A. Rücklé et al., "AdapterDrop: On the efficiency of adapters in transformers," in *Proc. Conf. Empirical Methods Natural Lang. Process.*, 2021, pp. 7930–7946.

[54] R. Wang et al., "K-adapter: Infusing knowledge into pre-trained models with adapters," in *Proc. Findings Assoc. Comput. Linguistics (ACL-IJCNLP)*, 2021, pp. 1405–1418.

[55] S. Chen et al., "AdaptFormer: Adapting vision transformers for scalable visual recognition," in *Proc. NIPS*, 2022, pp. 16664–16678.

[56] Z. Chen et al., "Vision transformer adapter for dense predictions," *Proc. Int. Conf. Learn. Represent.*, 2023, pp. 1–20.

[57] E. J. Hu et al., "LoRa: Low-rank adaptation of large language models," in *Proc. Int. Conf. Learn. Represent.*, 2022, pp. 1–13.

[58] S.-Y. Yeh, Y.-G. Hsieh, Z. Gao, B. B. W. Yang, G. Oh, and Y. Gong, "Navigating Text-To-Image customization: From LyCORIS fine-tuning to model evaluation," 2023, *arXiv:2309.14859*.

[59] S. Yang, W. Deng, M. Wang, J. Du, and J. Hu, "Orthogonality loss: Learning discriminative representations for face recognition," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 31, no. 6, pp. 2301–2314, Jun. 2021.

[60] Y. Wang et al., "Orthogonal deep features decomposition for age-invariant face recognition," in *Proc. Eur. Conf. Comput. Vis.*, Sep. 2018, pp. 764–779.

[61] C. Simon, P. Koniusz, R. Nock, and M. Harandi, "Adaptive subspaces for few-shot learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 4135–4144.

[62] K. Ranasinghe, M. Naseer, M. Hayat, S. Khan, and F. S. Khan, "Orthogonal projection loss," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 12313–12323.

[63] S.-A. Liu, Y. Zhang, Z. Qiu, H. Xie, Y. Zhang, and T. Yao, "Learning orthogonal prototypes for generalized few-shot semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 11319–11328.

[64] X. Sun et al., "FAIR1M: A benchmark dataset for fine-grained object recognition in high-resolution remote sensing imagery," 2021, *arXiv:2103.05569*.

[65] Y. Wu, K. Zhang, J. Wang, Y. Wang, Q. Wang, and X. Li, "GCWNet: A global context-weaving network for object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5619912.

[66] X. Huang, B. He, M. Tong, D. Wang, and C. He, "Few-shot object detection on remote sensing images via shared attention module and balanced fine-tuning strategy," *Remote Sens.*, vol. 13, no. 19, p. 3816, Sep. 2021.

[67] J. Pang, K. Chen, J. Shi, H. Feng, W. Ouyang, and D. Lin, "Libra R-CNN: Towards balanced learning for object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 821–830.

[68] B. Zhao, Y. Wu, X. Guan, L. Gao, and B. Zhang, "An improved aggregated-mosaic method for the sparse object detection of remote sensing imagery," *Remote Sens.*, vol. 13, no. 13, p. 2602, Jul. 2021.

[69] G. Ghiasi et al., "Simple copy-paste is a strong data augmentation method for instance segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 2918–2928.

[70] Y. Li, Q. Wang, X. Luo, and J. Yin, "Class-balanced contrastive learning for fine-grained airplane detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[71] X. Xie, G. Cheng, J. Wang, X. Yao, and J. Han, "Oriented R-CNN for object detection," in *Proc. IEEE Int. Conf. Comput. Vis.*, Oct. 2021, pp. 3500–3509.

[72] A. Aghajanyan, S. Gupta, and L. Zettlemoyer, "Intrinsic dimensionality explains the effectiveness of language model fine-tuning," in *Proc. 59th Annu. Meeting Assoc. Comput. Linguistics 11th Int. Joint Conf. Natural Lang. Process. (Long Papers)*, vol. 1, 2021, pp. 7319–7328.

[73] V. Lebedev, Y. Ganin, M. Rakhuba, I. V. Oseledets, and V. S. Lempitsky, "Speeding-up convolutional neural networks using fine-tuned cp-decomposition," in *Proc. Int. Conf. Learn. Represent.*, Y. Bengio and Y. LeCun, Eds., 2015, pp. 1–10.

[74] A.-H. Phan et al., "Stable low-rank tensor decomposition for compression of convolutional neural network," in *Proc. Eur. Conf. Comput. Vis.*, Aug. 2020, pp. 522–539.

[75] G. Cheng, J. Han, P. Zhou, and L. Guo, "Multi-class geospatial object detection and geographic image classification based on collection of part detectors," *ISPRS J. Photogramm. Remote Sens.*, vol. 98, pp. 119–132, Dec. 2014.

[76] M. Cramer, "The DGPF-test on digital airborne camera evaluation overview and test design," *Photogrammetrie Fernerkundung Geoinf.*, vol. 2010, no. 2, pp. 73–82, May 2010.

[77] K. Li, G. Wan, G. Cheng, L. Meng, and J. Han, "Object detection in optical remote sensing images: A survey and a new benchmark," *ISPRS J. Photogramm. Remote Sens.*, vol. 159, pp. 296–307, Jan. 2020.

[78] M. Everingham, L. Van Gool, C. K. I. Williams, J. Winn, and A. Zisserman, "The Pascal visual object classes (VOC) challenge," *Int. J. Comput. Vis.*, vol. 88, no. 2, pp. 303–338, 2010.

[79] L. van der Maaten and G. Hinton, "Visualizing data using t-SNE," *J. Mach. Learn. Res.*, vol. 9, pp. 2579–2605, Nov. 2008.



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